Topic Outline: Thesis Porsche Motorsport EMA3

1. Title

Development of a software analysis tool for MATLAB/Simulink models and their generated C code, providing a graphical signal dependency overview and computation data that can be used to optimize runtime and reduce cross-core data transitions on the Formula E VCU.

2. Background / Problem Statement

At Porsche Motorsport, a large part of the software deployed on the Formula E and LMDH vehicles is C code generated from MATLAB/Simulink models. These models are constantly updated and optimized by Porsche engineers to improve the vehicle's performance. If a certain function in one of these models needs to be changed, the engineer currently has to navigate through the model to track signal dependencies and understand the signal composition for all necessary inputs of the function to be improved. Tracing a signal through the model this way is time consuming and generates cognitive overhead for these engineers. The engineers need a quick overview of individual signal composition and the origin of computation.

3. Objective

The objective of this thesis is to develop a software analysis tool for MATLAB/Simulink models that accepts an SLX model or the generated code and artifacts from MATLAB Embedded Coder as input. Using a user-selected signal (either a block output or a signal in the model), the tool will perform backward tracing of this signal through the model. The data gained from this analysis will be used to create a structural calculation tree that shows all blocks, values, signals, and inputs that contribute to the selected signal. The result will be a readable map that shows all relationships of the selected signal, including additional information such as the origin of computation for each block. This gives engineers a fast way to understand, review, and document signal dependencies without manually navigating through the model.

4. Planned Approach

- Analysis of generated C code and artifacts from MATLAB Embedded Coder
- Development of a tool that performs signal correlation mapping and generates a dependency map
- Tool development with the support of AI coding models
- Optional: usage of extracted information to support multicore VCU code placement decisions

5. Use of AI Support

This thesis will make use of AI coding assistants GitHub Copilot

- AI tools are used to accelerate implementation of code
- No Porsche source code, or confidential data will be shared with external AI services.
- All AI generated code is reviewed, tested, and understood by the student before being integrated into the tool: the student remains fully responsible for correctness and code quality.
- The extent of AI assisted code generation is at the student's discretion but encouraged by the employer as part of AI assisted development workflow.
- Final tool architecture, design decisions, and validation are the student's own work.

6. Scope

- Core focus: development of the signal tracing / dependency mapping tool
- Multicore optimization recommendations are a supplementary, optional extension
- No automatic code replacement across cores; the tool provides recommendations only

7. Rough Timeline

- **Month 1:** Familiarization with MATLAB Embedded Coder artifacts and definition of tool architecture
- **Months 2:** Development of backward tracing logic and core analysis engine
- **Month 3:** Visualization / dependency map generation

- **Month 4:** Testing and validation against real Formula E models; optional multicore optimization extension
- **Month 5:** Debugging
- **Month 6:** Documentation and thesis write-up

8. Success Criteria

- The tool correctly traces signal dependencies on representative Formula E model examples
- Engineers can use the tool's output to understand signal composition faster than manual model navigation
- AI-assisted development workflow is documented as part of the thesis, providing the company with insight into its practical use and limitations

9. Support

Intern: Dr.-Ing. Stefan Lingl, Porsche EMA3
Extern: Prof. Dr.-Ing. Gernot Herbst, WHZ